

NeuroTune



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01

Music & the Brain



A brief overview of how music has been used, historically, as a means of communication.

Timeline



01 Prehistoric

02 Recent history

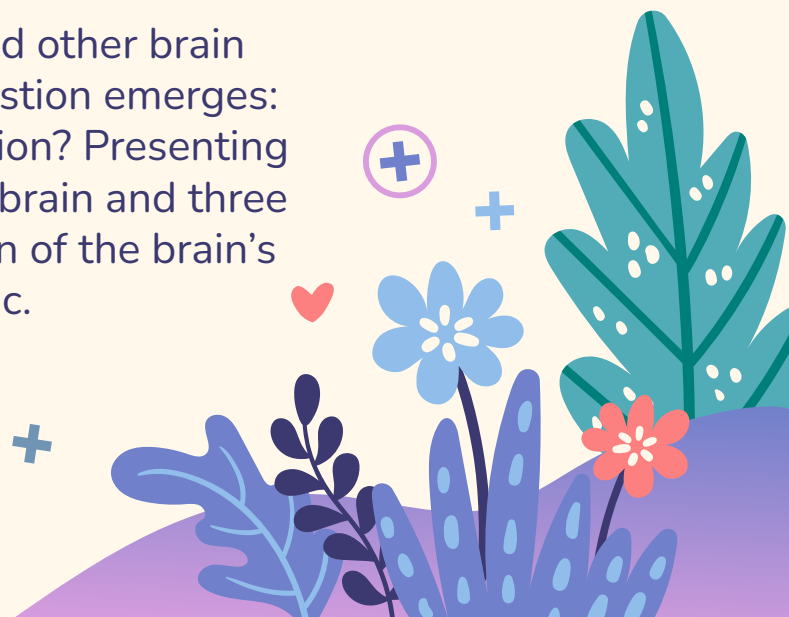
03 Modern Day

04 The Future



Abstract

With the introduction of EEG (and other brain monitoring technology) a new question emerges: What are the limits of communication? Presenting Neurotune - a bridge between the brain and three LEDs. An interactive approximation of the brain's interpretation of music.



Application - Human



Neurodivergence

Expression of feelings otherwise unheard.



Alternate Spoken Languages

Shared appreciation or dislike.



Small Children

Musical understanding even before speech.



Application - Animals



Communication

Turning feelings into understanding.



Music Therapy

Benefiting those whose feelings are overlooked.



Rehabilitation

Understanding the emotions behind actions.





02

Setup

Methodology

Code

01

Light up system

Input &
Output

02

Connect to EEG,
utilize

Collect

03

Data output from
brain activity

Analyze

04

What does this
mean?



Musical Analysis

Alpha Asymmetry

- ❑ Inverse

Right Hemisphere Association

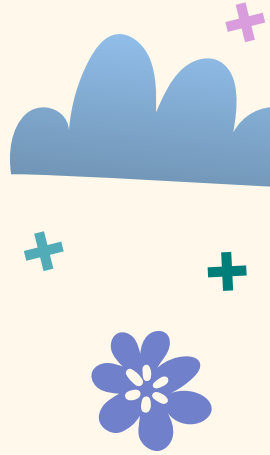
- ❑ Stronger when active listening
- ❑ Pitch & melodic Aspects

Approach-Withdrawal Model

- ❑ Anger
- ❑ Left vs Right

Left Hemisphere Association

- ❑ Temporal & Sequential (sound)
- ❑ Rhythm perception, discrimination



How The Code Works

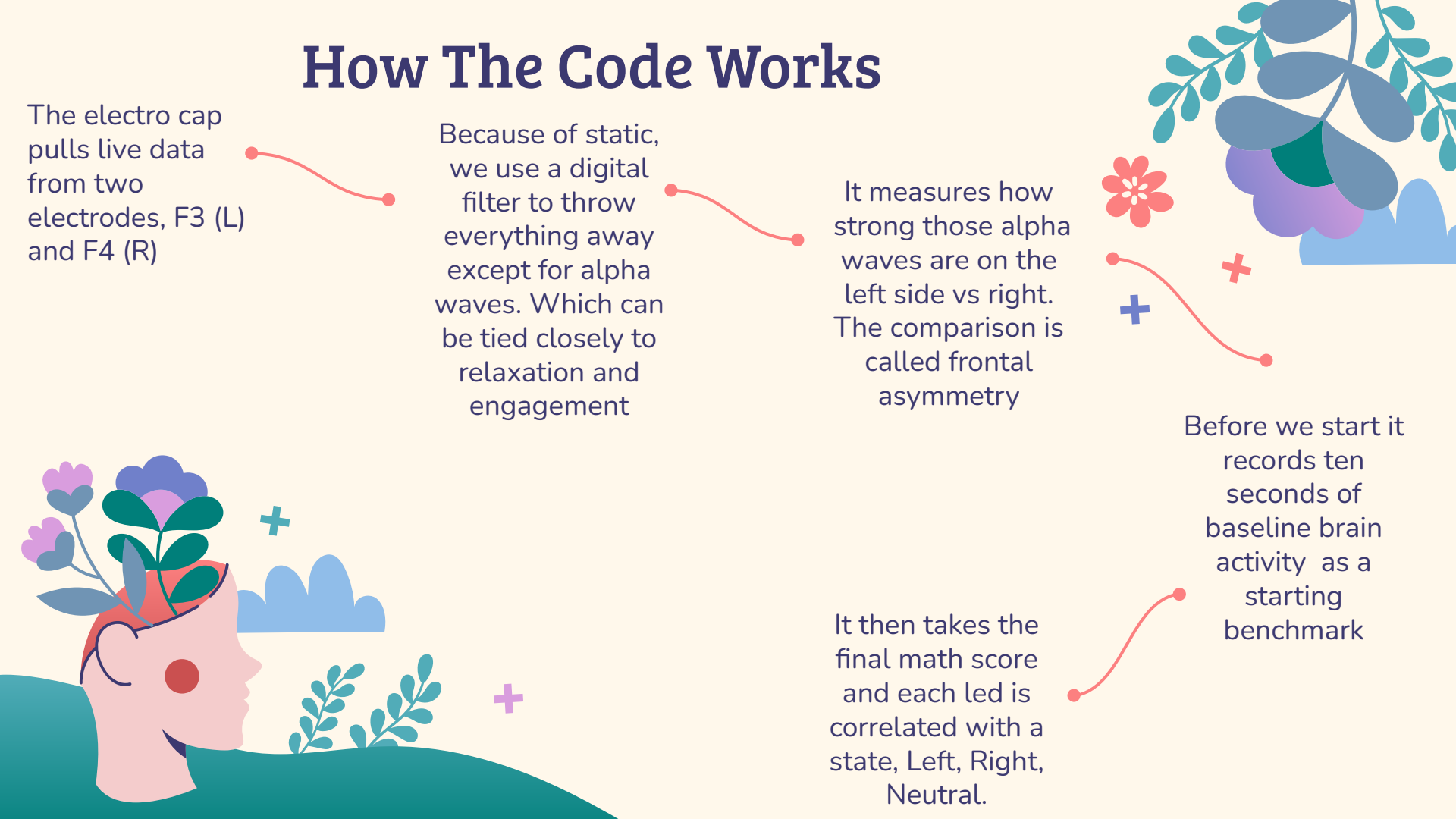
The electro cap pulls live data from two electrodes, F3 (L) and F4 (R)

Because of static, we use a digital filter to throw everything away except for alpha waves. Which can be tied closely to relaxation and engagement

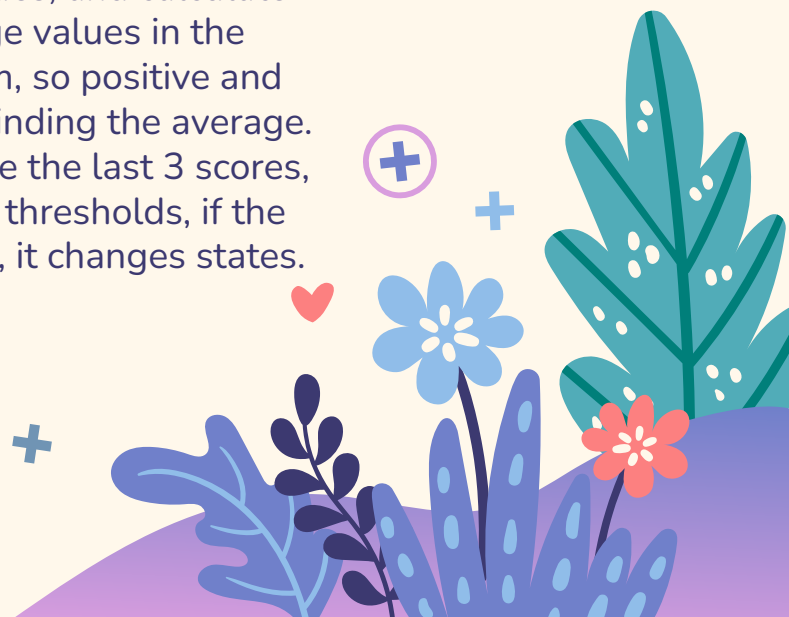
It measures how strong those alpha waves are on the left side vs right. The comparison is called frontal asymmetry

Before we start it records ten seconds of baseline brain activity as a starting benchmark

It then takes the final math score and each led is correlated with a state, Left, Right, Neutral.



Math!



First, we measure alpha wave strength. We take the filtered 8-12Hz brainwaves for both sides, and calculate their power, but taking all the voltage values in the currently time period and squares them, so positive and negative values both become positive. Finding the average. Then. Because brains flicker, we average the last 3 scores, so it doesn't flash wildly, we also have thresholds, if the final score has crossed a the trigger line, it changes states.

03 Findings: Live Demo



Analysis

Music to the Brain



Red Light

Right brain - Withdrawal



Yellow Light

Neutral



Green Light

Left Brain - Approach

Caveat



Equipment

EEG, Setup



Bias

Likes, Dislikes,
etc.



Methods

Monitoring,
Music



Environment

Presentation,
stress, other input

Recap



Abstract

NeuroTune



Setup

Methods + Research



Findings

Live Demo



Caveats

Constraints + other factors

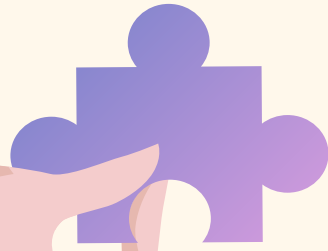




Importance of Simplicity

We wanted to come to the pitch with a final product, therefore we wanted to go with a strong conceptual idea, and create a demo using the a minimum viable product. It was important to us to not try to quantify music mathematically, but allow it to be a dynamic artform that could eventually be used as a means of communication.





Conclusion

Moving forward....



Sources

Frontal and parietal EEG alpha asymmetry: a large-scale investigation of short-term reliability on distinct EEG systems

From Neural Networks to Emotional Networks: A Systematic Review of EEG-Based Emotion Recognition in Cognitive Neuroscience and Real-World Applications

Asymmetric hemisphere activation in tenderness: evidence from EEG signals

How musical training affects cognitive development: rhythm, reward and other modulating variables

Live music stimulates the affective brain and emotionally entrains listeners in real time

Lateralization of auditory-cortex functions

The neuroscience of music perception: a narrative review

